Frontier



Ebb Carbon

R&D application

2026

General Application

The General Application applies to everyone; all applicants should complete this section

Public section

The content in this section (answers to questions 1(a) - (e)) will be made public on the Frontier GitHub repository after we announce funding decisions in the summer and fall. Include as much detail as possible but omit sensitive and proprietary information.

Organization name

Ebb Carbon, Inc.

Organization location (we welcome applicants from anywhere in the world)

South San Francisco, CA, USA

Name(s) of primary point(s) of contact for this application

Jeremy Loretz

Brief organization description, including organization type (e.g., for-profit, NGO, research institution/university) <20 words

Technology development and mCDR Supplier

1. Public summary of proposed project[1] to Frontier

a. **Project title** <20 words

Ocean Alkalinity Enhancement (OAE) Integrated with Desalination, Secondary Precipitation Risk Quantification

[1] We use “project” throughout this template, but the term is not intended to denote a single facility or experimental setup. The “project” being proposed to Frontier could include multiple facilities/locations for research activities.

b. **Project summary and field impact:** Provide an overview of your core research goal and the benefits you think it could provide to the broader CDR field. Tell us how your project meaningfully advances knowledge on at least one of the priority innovation areas identified in the RFP. What needs to happen for your project to unlock step-change advancements in cost and scale for this innovation area? Please include figures and system schematics, and be specific but concise. <1000 words

This project will build a foundation for economic scale-up of Ocean Alkalinity Enhancement (OAE) from pilots to large scale installations by mitigating the under-researched risk of secondary precipitation in real-world conditions.

The proposed research aims to evaluate the impacts on receiving waters of an OAE process integrated with an operational desalination facility. This project will leverage foundational data and deploy Ebb Carbon's technology to treat desalination brine, releasing aqueous alkalinity into the existing desalination plant's mixed effluent channel. This comprehensive effort will minimize secondary precipitation resulting from the increased alkalinity, while quantifying any remaining precipitation to be deducted as part of net Carbon Dioxide Removal (CDR) calculations.

Current major research around secondary precipitation and OAE can be found in academic publications including: Moras et al (2022, 2026), Hartmann et al (2023), and Suitner et al (2024, 2025). These focus on either global scale deployment or local high concentration alkalinity delivery, with efforts to determine thresholds and/or lag times appropriate to avoiding secondary precipitation triggered by alkalinity addition. These studies, along with others such as Ringham et al. (2024), Hashim et al. (2025), Valiero et al., (2024), represent bounding cases of OAE, but do not represent the reality of deployment today and for the coming decade. This is due to both scale mismatches on the amount of alkalinity added to a given intervention, and the difficulty in representing open system mixing, dilution, and advection in a laboratory or model context. In general, research on secondary precipitation to date has fallen into four categories: 1) natural analogs in environments including river outflows and specific oceanographic conditions (Wurgaft et al., 2016, 2021; Bustos-Serrano et al., 2009; Morse et al., 2003, 2007); 2) contained laboratory or mesocosm settings without the influence of tidal or other impacts on alkaline distribution; 3) early efforts to link laboratory conditions to the field by various air-sea exchange proxies (e.g., Moras et al., 2026), and 4) initial field observations that are yet-to-be-published (LOC-NESS, Project Macoma).

The significant knowledge gap between studies and deployment reality can be attributed to the absence of pragmatic specifications for commercially viable installations. Identifying and resolving the challenges today and near-future scales is essential for the commercialization and widespread adoption of OAE.

The most expedient route to achieving climate-relevant scales for OAE deployment involves integration with established industrial processes that utilize seawater, such as desalination. In combined processes, the produced alkalinity will interact with receiving water differently than hypothesized pure alkalinity additions to seawater, which is the commonly published mode. Unlike the dispersion of an alkalinity change over a large area of seawater, the alkalinity will mix with varied streams and be discharged through existing engineered infrastructure. Articulating these processes and their potential impacts on the risks of precipitation from changes in pH or alkalinity will guide future research to pose the most pertinent questions and concentrate on the practical needs at hand for precipitation triggers amongst OAE pathways. Collectively, these endeavors will facilitate OAE's

growth into a meaningful climate solution.

The research will lean on Ebb Carbon's multi-disciplinary team including ocean scientists, ocean modellers, chemical engineers, mechanical engineers, and project developers to innovate commercial scale deployments. This cross-linking will utilize existing knowledge from respective fields and targeted trial data to fuel effective implementations.

c. **Project objectives:** Describe your project in more detail: what are your key research questions and proposed approach for answering them, including methods, project location, and scale where applicable? Tell us why your approach is best-in-class, and how you're differentiated from any other organization working on a similar approach. Please include figures and system schematics, and be specific but concise. 1000-1500 words

Ebb Carbon's proven record of conducting world-class research in mCDR and pipeline of commercial projects will enable exemplary results in understanding secondary precipitation at industrial OAE deployments. Our novel process uses bipolar electrodialysis (BPED) to generate aqueous alkalinity from seawater. This process with its track record across various industries as a low cost and reliable means of producing low concentration caustic and acid, reduces technical risk in process scale-up. We have retired other key risks around permitting, process controls/safety/ and efficiency and have operated in a live ocean environment.

The key technical risks in OAE scaling are based around the alkalinity delivery method. Economically-viable alkalinity production must limit water waste and energy used during production, driving alkalinity concentrations that are elevated in pH and total alkalinity (TA) relative to ambient receiving water. This requires engineering mitigations to appropriately control alkalinity concentrations to limit precipitation of minerals that are already near or above saturation in seawater: aragonite, calcite, and brucite.

A combination of knowledge and data across laboratory, field, and modeling efforts uniquely positions Ebb Carbon to deliver valuable research in this field. The basis of the research proposed will stem from three primary assets to which Ebb Carbon has access.

1) Sustained academic and research partnerships including Pacific Northwest National Lab, University of Washington, NOAA Pacific Marine Environmental Laboratory, [C]Worthy, and Yale University. These partnerships are key for understanding phenomena at fundamental scales and adding scientific credibility and experience to frontier research in OAE.

2) Project Macoma, Ebb Carbon's fully integrated mCDR field research site in Port Angeles, WA. This site, operated since summer 2025, is the first Clean Water Act permitted mCDR deployment and the first permitted OAE site demonstrating an integrated desalination and electrochemical alkalinity generation system. Aqueous alkalinity was dispersed into open water and studied through various means over a period of six months, producing significant data on processes including secondary precipitation.

3) Commercial partners operating desalination processes, into which Ebb Carbon is currently working to integrate scientific research and OAE technology development. These connections allow in depth knowledge of implementation challenges and to quickly iterate to feasible solutions for rapid

deployment. This accelerates the ability to bring alkalinity sourcing and distribution towards annual kilotonne and megatonne scales.

Ebb Carbon's prime objective is to deliver mCDR at climate relevant scales, requiring accurate accounting for any deductions and uncertainties in alkaline delivery and carbon dioxide removal. The main area of focus for this research is to characterize the secondary mineral precipitation and dissolution that occurs under conditions of elevated alkalinity and pH at OAE dispersion points.

Secondary mineral precipitation following alkalinity release into seawater can have a multi-fold effect on high quality mCDR delivery quantification:

1. Loss of alkalinity due to predicted or measured non-reversible precipitation (e.g., carbonates) following release into seawater must be quantified, deducting alkalinity delivered from mCDR crediting analyses. With a risk of potential runaway reactions that lead to the further loss of alkalinity and dissolved inorganic carbon from receiving waters.
2. Loss of alkalinity due to predicted or measured reversible precipitation (e.g., brucite) following release into seawater must be quantified to ensure that alkalinity is redissolved and delivered in a way that promotes atmospheric carbon uptake.
3. Biological impacts to local ecosystems by mineral precipitation by way of increased turbidity, blocking light for photosynthesis within the water column or settling on aquatic vegetation. Potential impacts to predator-prey interactions or other ecosystem processes may impact the quality and value of the carbon removal.

To address this focus, three research questions and sub-questions are posed below to guide the proposed research and ensure actionable results. For each question an action plan is articulated.

1) **How will electrochemical OAE and desalination plants integrate?**
   What unit operations will each process possess and what will be shared infrastructure?
   At what steps do process streams diverge or merge, and what is the chemical species impact?
   How is each process improved or hindered through the integration?
   What co-benefits exist from the integration?
   Where can this be improved in the future?

   Ebb engineers will produce conceptual flow diagrams for various configurations of integrated electrochemical alkalinity generation and seawater desalination processes. These will highlight major constituents of the seawater, brine, alkaline, base, and other process streams as well as relative flow rates. From these constituents, we will model and quantify potential mineral formation risk at key process steps including: alkalinity introduction, mixing channel, and dispersion to receiving waters. These concepts will be compared to the design and operation of Project Macoma to demonstrate efficacy and impacts.

   The integrated plant concepts will also highlight the co-benefits of the installation to the desalination plant as well as additional revenue opportunities to the mCDR, allowing for further fundamental research around process technologies.

2) **What are the drivers for secondary mineral precipitation from OAE dispersal?**
   What has been observed and measured to date from existing field trials?
   What are the kinetic rates of precipitation and dissolution in a laboratory setting?
   What non-dimensional scaling factors apply to precipitation kinetics?

What is the morphology and composition of formed solids?
How does variation in salinity and temperature affect mineral formation?
Are there unanticipated risks of elevated alkalinity in existing desalination plants?

Project Macoma has already yielded major insights into mineral precipitation formation in an open system. These results need contextualization and further quantification, including additional geochemical modeling, prior to meaningful publication. Completing this body of work will move the current field forward by providing context and results from real-world, in-water precipitation dynamics, which are difficult to replicate in a laboratory setting.

In collaboration with research partners, high precision laboratory data studying the formation of precipitate under controlled conditions will be collated, compared to Project Macoma in-situ results, and published. These conditions will be heavily influenced by the findings of the first research question, and a detailed analysis of the various seawater and base product mixing modes trialed for the first time within Project Macoma that resulted in the release of high pH alkaline effluent with varying presence or absence of mineral precipitation.

Further reduced scale simulations of open releases will be studied in mesocosm settings and compared to observations at Project Macoma, allowing for advancement of modeling and monitoring techniques. Utilizing dimensionless number scaled analogs to simulate real deployments. We propose to conduct this work at PNNL-Sequim, continuing a long-standing research collaboration that has investigated precipitation thresholds in Salish Sea water using electrochemically-generated alkalinity.

Where possible, we would extend this research back into the field: PNNL-Sequim is in the process of updating permits for the on-site water treatment facility that has been used for previous, low pH alkaline releases (pH <9.0, Savoie et al., 2025). If permits move forward as expected, high pH alkaline releases (pH <12.0) will be conducted in late 2026, providing an opportunity for further in-water precipitation monitoring and modeling in a faster, tidally-influenced dilution regime. These releases will leverage additional high-resolution modeling assets including the Salish Sea Model and will provide additional validation data for PLUMES2.0. In addition, continued research is proposed to use the Project Macoma barge-based outfall for further OAE studies. Further data collection for precipitation dynamics and model validation is a key priority if this research moves forward in 2026.

Regardless of potential opportunities for expanding OAE research in the Salish Sea, integration risk assessment as part of implementation at commercial partner sites is the next target for research under this proposal. Ebb Carbon has partnered with the Saudi Water Authority (SWA) to advance OAE at a desalination facility. We will collect momentum and chemical data of existing mixed effluents to inform laboratory and model assessments of mixing alkalinity impacts, similar to work that we have done at PNNL-Sequim and Project Macoma. We will then proceed to pulse additions of alkalinity on site, quantifying the impacts of alkaline release on precipitation at high risk locations. This combined laboratory-model-field approach is necessary to advance understanding of precipitation in warm, salty waters influenced by suspended particulates, which has been understudied to date, particularly in the context of desalination mixing and outfall channels..

3) **What is needed for mCDR quantification modelling of an integrated process effluent?**
How does an engineered effluent dispersal apparatus change upscaling from near-field to

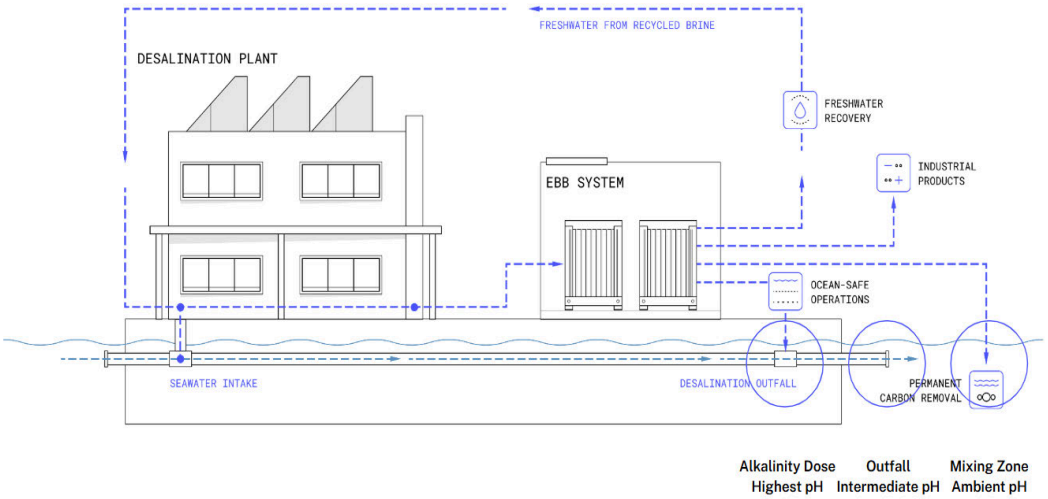
Regional Ocean Models (ROM)?
How does integrating OAE improve existing outfall characteristics at desalination plants?
What ROMs exist in locals of high desalination industry and what is required to make them suitable for mCDR quantification?
How to integrate biogeochemical physics into ROMs for prediction of precipitation rates?
What is the near-field pH/TA gradient at an integrated plant?

This is necessary for identifying the best suited application to mCDR quantification, combined with evaluation of high-resolution, near-field models from partner sites for applicability to the upscaling process. Knowledge gaps in OAE simulations will be analyzed in order to prioritize and implement additional for-purpose models, which will include precipitation and dissolution kinetics.

As part of this research, changes to the dynamics of effluent plumes (i.e., density shifts associated with the dilution or reduction of brine) will be studied and assessed for impacts on the desalination process and resulting positive or negative environmental impacts.

Desalination plumes can have significant negative impacts on local ecosystems due to high salinity, but the integration of OAE with desalination may bring effluents nearer to ambient salinities, and may improve local ecosystems due to local acidification mitigation.

Finally, Ebb Carbon scientists will integrate learnings from the first two research questions to inform model dynamics for predicting potential precipitation formation and dissolution. The ultimate goal of this research stage is to integrate and reduce uncertainty due to secondary precipitation and dilution processes with integration uncertainty in net carbon removal, while tracking and monitoring potential secondary impacts.



d. **Risks:** What are the biggest risks to your project's success and how will you mitigate those? Include scientific; technical; project execution; measurement, reporting and verification (MRV); ecosystem; financial; and any other risks. <500 words

A project of this scale and numerous partnerships has inherent risks to completion. Mitigating issues early and attentive management to these risks will give the project the greatest probability of success. Some of the major project risks are detailed below.

This project focuses on specific use cases and regional impacts, specifically targeting warm and salty waters in the context of desalination. While the results are likely generalizable to desalination integration or regional chemistry under similar climatologies, there can always be site specific impacts that will lead to different results and conclusions. One challenge in applying existing secondary precipitation research to desalination integration of OAE is that the majority of prior literature focuses on average surface seawater conditions (salinity ~35, temperature ~16-25°C, DIC and TA ~2000-2400 µmol/kg), and often in filtered seawater. Field data collected within this project will help to fill these knowledge gaps, with additional targeted laboratory research we will be able to robustly analyze field datasets in comparison to existing literature.

With the cross-discipline and cross-organizational intent of this project, there may become hurdles to sharing or releasing data. In one sense there is a timing risk present due to the slow peer-review process which predicates on consensus and thoroughness versus speed. There is also a risk of individual parties having hesitancy to release data. This can be due to either preference for different publications from academic partners, or from sensitivity for proprietary and/or security reasons with commercial partners. To mitigate this collaboration, agreements and expectations will be discussed and agreed at the onset of the project, and reaffirmed prior to periods of public disclosure.

The future availability of research assets at Project Macoma is uncertain at present. Ebb Carbon has owned and operated it to date, but due to a series of externalities preventing continuance we are actively engaging a research entity to take over scientific studies at the site. Successful handover will allow for additional and complimentary study at a well baselined field site. In the event operations must fully subside, this proposed research can still be successful relying on the other assets and existing data.

As is common in large projects, timelines can always shift due to unexpected events. The largest variance in timing for this project is expected to come from onsite activities at a partner site. While there is a pipeline of options available today, finalizing business and legal agreements and then commencing operations are always long duration tasks. In the absence of a site in time, this project will shift resources to the other available assets and use generalized conditions.

Finally, this project is relying on global partners and operators which can lead to significant project risk with the ever changing geopolitical sphere. Direct effects could involve researcher travel, site operations, material procurement, or data access. These may have financial, schedule, or execution impacts. To mitigate these risks the project plan may need to adjust in real time and a continued progress conversation with Frontier is expected.

e. **Proposal to Frontier:** Please list your primary research deliverables, project timeline, and requested funding below. If you are selected for Frontier funding, this table will form the basis of contract discussions.

Primary research deliverables	1. Peer-reviewed publication(s) of Project Macoma precipitation observations, with

(List tangible outputs of the project: e.g., validated datasets, peer-reviewed models, bench-scale prototypes, or field trial results. Should focus on resolving a specific bottleneck or uncertainty identified in Section 1.a)	conference follow-up. 2. Technical report on OAE and desalination integration. 3. Technical report on precipitation uncertainty modelling results. 4. Peer-reviewed publication on laboratory precipitation experiments, with conference follow-up. 5. Technical report on precipitation threshold recommendations at elevated salinity and temperature. 6. Technical report on field trial results at commercial field site. 7. Peer-reviewed publication(s) on OAE and desalination integration risk assessment, with conference follow-up.
Field impact *(Briefly describe how these deliverables resolve key bottlenecks for a specific CDR pathway or broader CDR field, e.g. efficiency gains, scaling unlocks, or reducing measurement uncertainty.)*	This work is aimed at addressing two major bottlenecks for the mCDR field: 1. Minimization and quantification of secondary mineral precipitation due to increased alkalinity and pH of effluents. This work will both add to the field and better address precipitation-related concerns by determining reasonable values for deduction in net CDR calculations, showing low probability of runaway precipitation events, and quantifying environmental impact. Further this work will allow other researchers to frame their efforts in terms of real deployments rather than conceptual thought experiments, which is particularly important when moving from the laboratory to the field given the impact of momentum and mixing on dilution and precipitation of alkalinity. 2. Increased commercialization of mCDR technologies to meet growing environmental needs. It is well shown that having multiple benefits including both environmental and economic is required for mCDR technologies to prosper. Through these studies we will further demonstrate the multitude of benefits from the Ebb Carbon process including: CDR, increased drinking water production, critical mineral extraction, and

	low-carbon chemical production.
Project timeline & key milestones *(At what point should Frontier consider the research project complete? Include major progress markers. Frontier generally prefers projects with clear readouts within ~18 months.)*	The full scope of this project is expected to take 1 year and be completed by June of 2027. Milestones are listed below: 1. Publication of Project Macoma results (Sept 2026) 2. OAE and desalination process modelling (Nov 2026) 3. ROM of combined desalination and OAE effluents (Dec 2026) 4. Ex-situ mineral precipitation characterization (Nov 2026) 5. In-situ mineral precipitation characterization at commercial plant (March 2027) 6. Integration risk assessment and mitigation strategy (May 2027)
Total grant request (US$) *(The total funding required to execute the research scope described above. Frontier R&D grants typically range from US$250k–$750k.)*	$750k